The Link-Lift Spillover Obstruction for a Single-Star Repair

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Abstract

We lift a centered six-plaquette repair to the links of the open 2^4 block and compute the least external curvature it must create.

Let A map noncentral link variations to the five centered incident-curvature modes and let B map them to the other 210 plaquettes. Define

$$H = R^*R, \qquad Ry = \operatorname*{argmin}_{Ax=y} \|Bx\|.$$

Theorem 1. *For each of the eight six-plaquette interior links, the spectrum of H is*

$$\{0,\ 1.121212\ldots,\ 1.121212\ldots,\ 1.635983\ldots,\ 1.635983\ldots\}.$$

Thus only one centered mode has a zero-spill lift; four modes necessarily export curvature.

Proof. Choose an orthonormal basis of the centered star. A particular right inverse is $L_0 = A^*(AA^*)^{-1}$. Varying it by $\ker A$ shows that the minimum external field is the orthogonal residual of BL_0 modulo $B\ker A$. The displayed finite matrix R^*R is diagonalized by the executable certificate; all eight incidence stars are congruent. □

After action drop, fifteen-dimensional Jacobian, Haar loss, and local comparison loss are combined, the worst-mode single-star repair misses the required free energy even assuming zero external curvature before repair.

Remark 1. *The ideal Stage 43 repair therefore does not lift locally. A coordinated polymer repair must reduce the effective spillover spectrum.*